

Sample Midterm 2, MATH 4211/6211 Optimization

NAME: _____ ID#: _____ Section: 4211/6211

Show all your work. Answers without solution steps will not receive credits.

1. (4 points) Consider the problem of minimizing $f(x) = x^{4/3} = (\sqrt[3]{x})^4$, $x \in \mathbb{R}$. Note that 0 is the global minimizer of f .
- (a) Write down the algorithm for Newton's method applied to this problem.
 - (b) Show that as long as the starting point is not 0, the algorithm in part (a) does not converge to 0 (no matter how close to 0 we start).

Solution.

- (a) We have $f(x) = x^{4/3}$, $f'(x) = (4/3)x^{1/3}$, $f''(x) = (4/9)x^{-2/3}$. So the Newton's method yields the following algorithm:

$$x^{(k+1)} = x^{(k)} - \frac{f'(x^{(k)})}{f''(x^{(k)})} = x^{(k)} - 3x^{(k)} = -2x^{(k)}$$

- (b) If $x^{(0)} \neq 0$, then the algorithm above yields

$$x^{(k)} = -2x^{(k-1)} = \dots = (-2)^k x^{(0)}$$

which does not converge to 0 as $k \rightarrow \infty$.

2. (4 points) Suppose we apply a fixed step size gradient algorithm to minimize

$$f(\mathbf{x}) = \mathbf{x}^T \begin{bmatrix} 3/2 & 2 \\ 0 & 3/2 \end{bmatrix} \mathbf{x} + \mathbf{x}^T \begin{bmatrix} 3 \\ -1 \end{bmatrix} - 22$$

- (a) Find the range of values of the step size for which the algorithm converges to the minimizer.
- (b) Suppose we use a step size of 1000 (which is too large). Find an initial condition that will cause the algorithm to diverge (not converge).

Solution.

- (a) The function $f(\mathbf{x})$ can be rewritten in the standard form

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \begin{bmatrix} 6 & 4 \\ 4 & 6 \end{bmatrix} \mathbf{x} + \mathbf{x}^T \begin{bmatrix} 3 \\ -1 \end{bmatrix} - 22$$

Denote

$$\mathbf{Q} = \begin{bmatrix} 6 & 4 \\ 4 & 6 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

Solving λ from $\det(\lambda \mathbf{I} - \mathbf{Q}) = 0$, we obtain the eigenvalues of $\mathbf{Q}$ as $\lambda = 2$ and $\lambda = 10$. Therefore the range of values of the fixed step size is $(0, 2/10)$.

- (b) We only need to set $\mathbf{x}^{(0)}$ such that $\mathbf{x}^{(0)} - \mathbf{x}^*$ is the eigenvector corresponding to the largest eigenvalue 5 of $\mathbf{Q}$. To this end, we can compute $\mathbf{x}^* = \mathbf{Q}^{-1} \mathbf{b} = [11/5, -9/5]^T$, and an eigenvector of $\mathbf{Q}$ corresponding to $\lambda = 5$ as $\mathbf{v} = [1, 1]^T$. Then with $\mathbf{x}^{(0)} = \mathbf{x}^* + \mathbf{v} = [16/5, -4/5]^T$, the gradient descent algorithm with fixed step size $\alpha = 1000$ diverges.

3. (4 points) Given $f : \mathbb{R}^n \rightarrow \mathbb{R}$, consider the general iterative algorithm

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha_k \mathbf{d}^{(k)},$$

where $\mathbf{d}^{(1)}, \mathbf{d}^{(2)}, \dots$ are given vectors in $\mathbb{R}^n$, and α_k is chosen to minimize $f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)})$; that is,

$$\alpha_k = \operatorname{argmin} f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}).$$

Show that for each k , the vector $\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}$ is orthogonal to $\nabla f(\mathbf{x}^{(k+1)})$ (assuming the gradient exists).

Proof. If $\alpha_k = 0$, then $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)}$ and hence $\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)} = 0$ which is orthogonal to any vector. Now suppose $\alpha_k \neq 0$. Since $\alpha_k = \operatorname{argmin} f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)})$, by FONC, there is

$$0 = \nabla f(\mathbf{x}^{(k)} + \alpha_k \mathbf{d}^{(k)})^T \mathbf{d}^{(k)} = \nabla f(\mathbf{x}^{(k+1)})^T \mathbf{d}^{(k)} = \frac{1}{\alpha_k} \nabla f(\mathbf{x}^{(k+1)})^T (\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)})$$

Therefore $\nabla f(\mathbf{x}^{(k+1)})$ and $\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}$ are orthogonal.

4. (4 points) Suppose $\mathbf{Q}$ is symmetric and positive definite. Show that if $\{\mathbf{d}^{(1)}, \dots, \mathbf{d}^{(n)}\}$ is a $\mathbf{Q}$ -conjugate set that is also orthogonal (i.e., $\mathbf{d}^{(i)T} \mathbf{d}^{(j)} = 0$ for all $i, j = 1, \dots, n$ and $i \neq j$), and $\mathbf{d}^{(i)} \neq \mathbf{0}$, $i = 1, \dots, n$, then each $\mathbf{d}^{(i)}$, $i = 1, \dots, n$, is an eigenvector of $\mathbf{Q}$.

Proof. Since $\{\mathbf{d}^{(1)}, \dots, \mathbf{d}^{(n)}\}$ is a $\mathbf{Q}$ -conjugate, we know they are linearly independent (and hence form a basis of $\mathbb{R}^n$). For any i , since $\mathbf{Q}\mathbf{d}^{(i)} \in \mathbb{R}^n$, we know there exist $\beta_1, \dots, \beta_n \in \mathbb{R}$ such that

$$\mathbf{Q}\mathbf{d}^{(i)} = \beta_1 \mathbf{d}^{(1)} + \dots + \beta_n \mathbf{d}^{(n)}$$

Now we show $\mathbf{Q}\mathbf{d}^{(i)} = \beta_i \mathbf{d}^{(i)}$. To this end, we pre-multiply $\mathbf{d}^{(j)T}$ where $j \neq i$ to both sides above and obtain $0 = \beta_j \|\mathbf{d}^{(j)}\|^2$ since all $\mathbf{d}^{(i)}$ are orthogonal and $\mathbf{Q}$ -conjugate. Since $\mathbf{d}^{(j)} \neq \mathbf{0}$, we have $\beta_j = 0$.

Therefore $\mathbf{d}^{(i)}$ is an eigenvector of $\mathbf{Q}$ (with eigenvalue β_i). Since i is arbitrary, we know every $\mathbf{d}^{(i)}$ is an eigenvector of $\mathbf{Q}$.

5. (4 points) Consider the quadratic function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ given by

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} - \mathbf{x}^T \mathbf{b},$$

where $\mathbf{Q}$ is symmetric and positive definite. Let $\mathbf{D} \in \mathbb{R}^{n \times r}$ be of rank r , and $\mathbf{x}_0 \in \mathbb{R}^n$. Define the function $\phi : \mathbb{R}^r \rightarrow \mathbb{R}$ by

$$\phi(\mathbf{a}) = f(\mathbf{x}_0 + \mathbf{D}\mathbf{a}).$$

Show that ϕ is a quadratic function with a positive definite quadratic term.

Proof. We first compute and simplify $\phi(\mathbf{a})$:

$$\phi(\mathbf{a}) = f(\mathbf{x}_0 + \mathbf{D}\mathbf{a}) = \frac{1}{2} \mathbf{a}^T \mathbf{D}^T \mathbf{Q} \mathbf{D} \mathbf{a} + (\mathbf{D}^T \mathbf{Q} \mathbf{x}_0 + \mathbf{D}^T \mathbf{b})^T \mathbf{a} + \frac{1}{2} \mathbf{x}_0^T \mathbf{Q} \mathbf{x}_0 - \mathbf{x}_0^T \mathbf{b}$$

Therefore, $\phi(\mathbf{a})$ is quadratic in $\mathbf{a}$ with symmetric matrix $\mathbf{D}^T \mathbf{Q} \mathbf{D}$ in the quadratic term.

To show $\mathbf{D}^T \mathbf{Q} \mathbf{D}$ is positive definite, we first note that $\mathbf{a}^T \mathbf{D}^T \mathbf{Q} \mathbf{D} \mathbf{a} = (\mathbf{D}\mathbf{a})^T \mathbf{Q} (\mathbf{D}\mathbf{a}) \geq 0$ for any $\mathbf{a} \in \mathbb{R}^r$ since $\mathbf{Q}$ is positive definite. Moreover,

$$\mathbf{a}^T \mathbf{D}^T \mathbf{Q} \mathbf{D} \mathbf{a} = 0 \iff \mathbf{D}\mathbf{a} = \mathbf{0} \iff \mathbf{a} = \mathbf{0}$$

where the first equivalency is due to the positive definiteness of $\mathbf{Q}$, and the second equivalency is due to the fact that $\mathbf{D}$ has rank r (which also means $r \leq n$).